SECJ2203: Software Engineering

Project Proposal

KADA Online System

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1. Introduction

1.1 The Goal

The Koperasi Kakitangan KADA Kelantan Berhad is a cooperative under the Lembaga Kemajuan Pertanian Kemubu (KADA) that provides its members with several services, such as financing services and member welfare programs. Currently, all the cooperative processes are managed manually. However, manual operations may cause several challenges, such as time-consuming and human error.

The main goal of this project is to develop a standalone, web-based digital system to automate the cooperative processes. The proposed system will ease the cooperative process, and increase efficiency.

The objectives of this project include:

i. **To develop an automated member registration and loan application system** that digitalises member registration and loan application process.
ii. **To enhance service accessibility of the cooperative members** by creating an online platform that enables the members to view and manage their profile and apply for loan easily from any location.
iii. **To ensure data security and confidentiality of the cooperative** by implementing security features, such as user authentication and encryption to prevent unauthorized access of members' personal and financial data.
iv. **To develop an automated report generating system** that could generate financial reports based on the cooperative database.

1.2 The Scope

For the development of this proposed system, several technologies will be used to design and implement an efficient, secure, and user-friendly system. These technologies include:

i. Online Web Hosting
ii. mySQL Database
iii. HTML Technology
iv. CSS Technology
v. JavaScript Technology
vi. PHP Technology

Table 1.1 shows the technology and its description to be used to develop this system project.

No.	Technology	Description
1.	Online Web Hosting	Host the system on a cloud platform, ensuring high availability and remote access for members and admins.
2.	mySQL Database	Store and manage the cooperative data, including members and loan applicants data, providing secure access to information.
3.	HTML Technology	Build the structure of the web pages, forming the base layout and content organization for the system interface.
4.	CSS Technology	Add design to the interface to provide a visually appealing and user-friendly layout.
5.	JavaScript Technology	Improve user experience by adding functionalities such as form validation and real-time updates.
6.	PHP Technology	Manage backend operations, including data processing, user authentication, and communication with the database.

The proposed KADA Online System contains various module including:

1. Log In
2. Apply Membership
3. Apply Loan
4. View Loan Application Status
5. Update User Profile
6. View Financial Status
7. View List
8. Approve Membership
9. Approve Loan Application
10. Record Transaction
11. Update Policies
12. Update Banner
13. Generate Monthly and Annual Report

Table 1.2 shows the system modules and its description

No.	Module	Description
1.	Log In	Members and administrators can log in to the system using their username and password.
2.	Apply Membership	No-members need to apply for membership by filling the new member application form.
3.	Apply Loan	Members can apply for a loan by filling the loan application form.
4.	View Loan Application Status	The loan applicants can view their application status (pending, approved, or reject) on the member interface.
5.	Update User Profile	Members can update their personal information and the family and inheritor information through the Kemaskini function in the member interface.
6.	View Financial Status	Members can view their financial status including the loan and share information in the member interface.
7.	View List	Admins can view member and loan applicant lists in the admin interface.
8.	Approve Membership	Admins can approve membership applications of new users in the admin interface.
9.	Approve Loan Application	Admins can approve loan applications of members in the admin interface.
10.	Record Transaction	Admins can record the transaction records and update financial status such as the amount of share capital and loan.
11.	Update policies	Admins can update policies such as the minimum membership fees and the financing rates in the admin interface.
12.	Update banner	Admins can update advertising banners in the admin interface of the system.
13.	Generate Monthly and Annual Report	Admins can generate and print monthly and annual financial reports from the admin interface.

2. Software Process Model

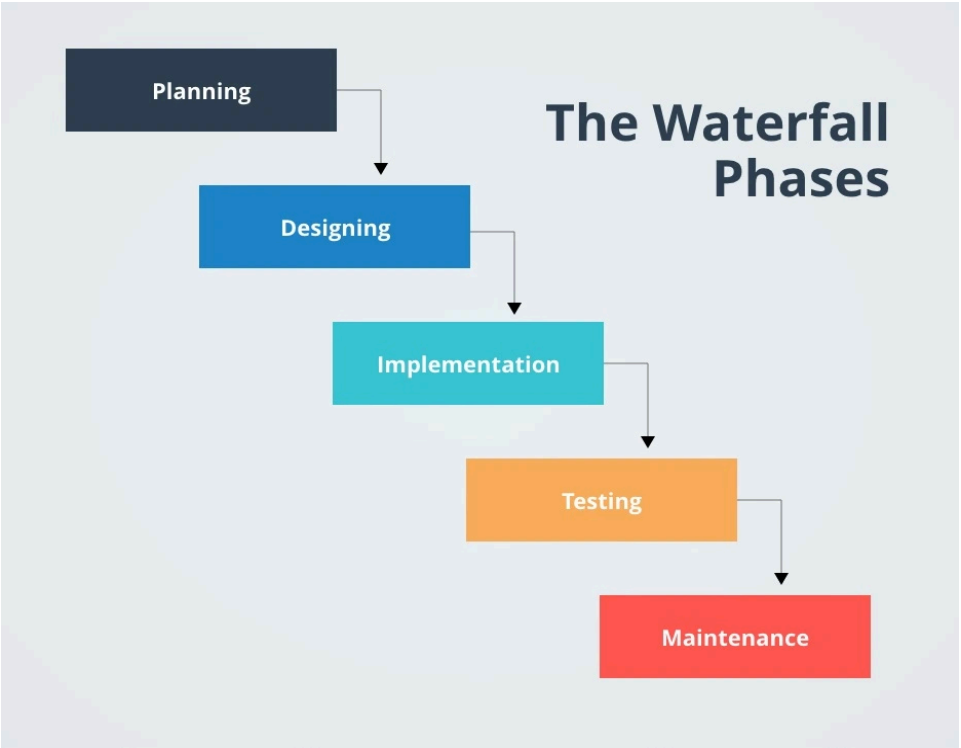
2.1 The Goal

A software process model is an organized method for planning and managing tasks in order to efficiently create high-quality software. An organized approach to development is ensured by a software process model, which specifies a list of actions, required inputs and outputs, preconditions, and postconditions. It offers a road map for guiding development teams, improving teamwork, and making it easier to accomplish the project's primary objectives.

2.2 The Chosen Software Process Model

For this system, the **waterfall model** is suitable as a software process model because it is easy to manage where it allows for a structured, sequential approach where each phase has specific deliverables before moving to the next. This ensures a well-documented and systematic development process.

Diagram 2.1 The Waterfall Phases



2.3 Project Duration

Phase 1: Analysis (Duration: 3 weeks)

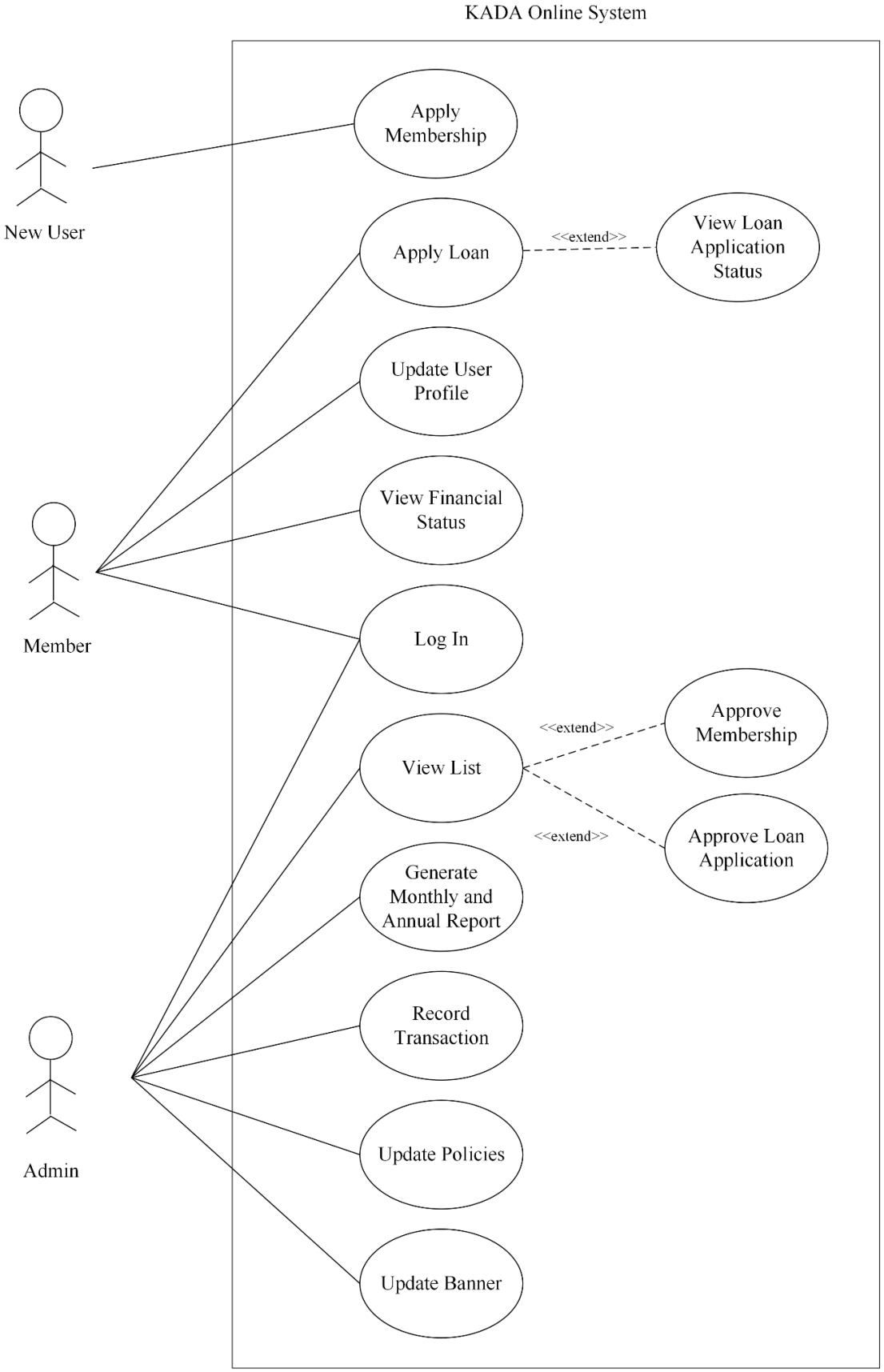
In this phase, the main focus is on collecting both functional and non-functional requirements for the system. The process begins with an online meeting session between Encik Ahmad from the cooperative and all members to define the system scope and objectives. During this

discussion, we identify the main features of the system, such as member application, loan application and financial report.

Our group leader, Yi Ya leads the requirement gathering process, ensuring that all necessary information is obtained through WhatsApp group discussion with the cooperative. Based on the discussion, we prepare a comprehensive software requirements specification (SRS), which details all functional requirements and non-functional requirements. Jia Lin is responsible for evaluating the current system, identifying weaknesses or errors, and drafting new requirements for the improved system. Additionally, use case diagrams are created to visually represent system interactions, aiding the stakeholder in understanding the proposed system structure. These deliverables form the basis of the following design phases.

Use Case Diagram

Diagram 2.2 Use Case Diagram



Use Case 1: Apply Membership

Table 2.1 Use Case Description for Apply Membership

Use Case Name:	Apply Membership
Description:	This use case allows new users to submit required information to apply for membership in the cooperative.
Primary Actor:	New Users
Secondary Actor:	Admin
Include Use Cases:	-
Preconditions:	1. User is an existing employee at KADA. 2. User has not registered for membership at the cooperative before.
Main Flow:	1. User opens up the KADA web application and the log in page shows up upon the first entry. 2. User clicks on the prompt at the bottom of the login screen "Daftar Ahli Baharu". 3. The user is directed to the Registration page. 4. User provides necessary information and uploads the required document to continue.
Postconditions:	1. Upon submission, the system shows status as "Sedang Diproses".

Use Case 2: Log In

Table 2.2 Use Case Description for Log In

Use Case Name:	Log In
Description:	This use case allows existing users to login after typing in the username and correct password.
Primary Actor:	Member
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. User must be an existing member in the cooperative. 2. User must type the correct ID and password

Main Flow:	1. User opens up the KADA web application and is directed to the login page. 2. User key in the correct user ID and password in their respective column. 3. User clicks “Log masuk”.
Postconditions:	1. The system directs to the main page.

Use Case 3: Apply Loan and View Loan Application Status

Table 2.3 Use Case Description for Apply Loan and View Loan Application Status

Use Case Name:	Apply Loan and View Loan Application Status
Description:	This use case allows a member to apply a loan application and track its status through the system.
Primary Actor:	Member
Secondary Actor:	Admin
Extend Use Cases:	1. View loan application status
Preconditions:	1. The member must be logged into the system.
Main Flow:	1. The member navigates to the loan application page and enters loan details. 2. The system validates the application information for completeness and accuracy. 3. Upon successful validation, the system saves the application with “Sedang diproses” status. 4. The system displays a confirmation message to the member, indicating that the application has been submitted. 5. The board of directors reviews the loan application. ● If approved, the system updates the application status to “Diterima.” ● If not approved, the system updates the application status to “Ditolak.” 6. The member accesses the loan status page to view the current status of their application. 7. The system displays the loan application status and any additional relevant information.
Postconditions:	1. The loan application status is saved with an initial “Sedang diproses” status and viewable by the member. 2. The application status is updated to “Diterima” if the board of directors approves it or to “Ditolak” if it is not approved.

Use Case 4: Update User Profile

Table 2.4 Use Case Description for Update User Profile

Use Case Name:	Update User Profile
Description:	Members can update their personal information and the family and inheritor information through the Update function in the member interface of the system.
Primary Actor:	Member
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The member must be logged into the system.
Main Flow:	1. The member navigates the Profil page. 2. The member chooses to update their personal information or their family and inheritor information. 3. The member presses on the "Kemaskni" button. 4. The member updates their information. 5. The member presses on the "Simpan" button to save their information. 6. The system updates the database. 7. The system prompts a confirmation message. 8. The member presses "OK". 9. The member redirects to the "Profil" page.
Postconditions:	1. The information updated is saved.

Use Case 5: View Financial Status

Table 2.5 Use Case Description for View Financial Status

Use Case Name:	View Financial Status
Description:	Members can view their financial status including the loan and share information.
Primary Actor:	Member
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The member must be logged into the system.
Main Flow:	1. The user navigates the "Laman Utama". 2. The user view the "Maklumat Saham Ahli" and "Maklumat Pinjaman Ahli".

Postconditions:	-

Use Case 6: Approve Membership

Table 2.6 Use Case Description for Approve Membership

Use Case Name:	Approve Membership
Description:	This use case allows admin to approve the new membership applications.
Primary Actor:	Admin
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The admin must be logged in with appropriate authorisation.
Main Flow:	1. Admin navigates to “Permohonan Anggota”. 2. The admin clicks “...” to view more about the applications. 3. The admin reviews the membership applications. ● If approved, the system updates the application status to “Diterima.” ● If rejected, the system updates the application status to “Ditolak.”
Postconditions:	1. The system is updated to “Diterima” status if approved or “Ditolak” status if rejected. 2. The users can view their applications status. 3. When the membership applications’ status is updated to “Diterima” status, the application will be updated to “Senarai Anggota”.

Use Case 7: Approve Loan Application

Table 2.7 Use Case Description for Approve Loan Application

Use Case Name:	Approve Loan Application
Description:	This use case allows admin to approve the new membership applications.
Primary Actor:	Admin
Secondary Actor:	Member
Include Use Cases:	-
Preconditions:	1. The admin must be logged in with appropriate authorisation.

Main Flow:	1. Admin navigates to “Permohonan Pinjaman”. 2. The admin clicks “...” to view more about the applications. 3. The admin reviews the loan applications. ● If approved, the system updates the application status to “Diterima.” ● If rejected, the system updates the application status to “Ditolak.”
Postconditions:	1. The system is updated to “Diterima” status if approved or “Ditolak” status if rejected. 2. The members can view their applications status. 3. When the loan applications’ status is updated to “Diterima” status, the application will be updated to “Senarai Peminjam”.

Use Case 8: View List

Table 2.8 Use Case Description for View List

Use Case Name:	View List
Description:	This use case allows the admin to view the list of member applications, loan applications, existing members and loans through the system.
Primary Actor:	Admin
Secondary Actor:	Member
Extend Use Cases:	1. Approve Membership 2. Approve Loan Applications
Preconditions:	1. The admin must be logged in with appropriate authorisation.
Main Flow:	1. The admin navigates the “Permohonan Anggota” and “Permohonan Pinjaman” tabs to view the membership and loan applications. 2. The admin navigates to “Senarai Anggota” and “Senarai Peminjam” to view the approved applications, existing members and loans.
Postconditions:	-

Use Case 9: Record Transaction

Table 2.9 Use Case Description for Record Transaction

Use Case Name:	Record Transaction

Description:	This use case allows transactions that may affect the user financial status to be updated. This includes updating the amount of share capital, funds, contributions and loans.
Primary Actor:	Admin
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The admin must be logged in with appropriate authorisation. 2. The transaction between the cooperative and the member is settled.
Main Flow:	1. The admin navigates to the Record Transaction tab. 2. The system displays the list of members. 3. The admin searches and selects the member to be updated. 4. The system displays the financial status of the member. 5. The admin enters the amount to be added or subtracted. 6. The system prompts a confirmation message. 7. The admin clicks "Ok". 8. The system updates the database and the transaction is recorded.
Postconditions:	-

Use Case 10: Update Policies

Table 2.10 Use Case Description for Update Policies

Use Case Name:	Update Policies and Banner
Description:	This use case allows the admin to update the policies such as the minimum membership fee, minimum share capital, financing rates, maximum loan repayment period and maximum loan allowed.
Primary Actor:	Admin
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The admin must be logged in with appropriate authorisation.
Main Flow:	1. The admin navigates to "Kemaskini Polisi". 2. The system displays the list of limits and rates. 3. The admin clicks "Kemaskini" to update the relevant section. 4. The system updates the new values. 5. The system prompts a confirmation message.

	6. The admin clicks “Ok”.
Postconditions:	The financial status of the member is updated and the transaction is logged.

Use Case 11: Update Banner

Table 2.11 Use Case Description for Update Banner

Use Case Name:	Update Banner
Description:	This use case allows the admin to update the banner displayed in the main page of the member view.
Primary Actor:	Admin
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The admin must be logged in with appropriate authorisation.
Main Flow:	1. The admin navigates to “Kemaskini Banner”. 2. The admin uploads a new photo. 3. The admin clicks “Kemaskini”. 4. The system prompts a confirmation message. 5. The admin clicks “Ok”.
Postconditions:	The banner is updated.

Use Case 12: Generate Monthly and Annual Financial Reports

Table 2.12 Use Case Description for Generate Monthly and Annual Financial Reports

Use Case Name:	Generate Monthly and Annual Financial Reports
Description:	This use case allows the generation and viewing of monthly and annual financial reports by an authorized user. These reports include data regarding new member and loan applications, the amount of fees and contributions claimed during the current year.
Primary Actor:	Admin
Secondary Actor:	-
Include Use Cases:	-
Preconditions:	1. The admin must be logged in with appropriate authorisation. 2. Financial data must be up-to-date in the system.

Main Flow:	1. The admin navigates to the Financial Reporting section. 2. The admin selects the report type, such as monthly or annual and specifies the time frame. ● If the admin selects an invalid time frame, the system prompts to correct the input before proceeding with report generation. 3. The system retrieves relevant data from the database, including new member applications, loan applications, fees claimed and contributions claimed. 4. The system generates a financial summary and detailed breakdown of each category for the specified period. 5. The system displays the report, providing options to view in detail, export or print the report.
Postconditions:	1. The system generates a comprehensive monthly or annual financial report based on the selected time frame and data parameters. 2. The report is viewable by the admin with options to export or print.

Phase 2: Design (Duration: 4 weeks)

This phase consists of two main components: High-Level Design (HLD) and Low-Level Design (LLD). HLD provides an overview of the system architecture, outlining the various components, including hardware, software interfaces and user interactions. Jie Ying leads this phase, ensuring that the design aligns with our business requirements and establishes guidelines for programmers to help them understand the system implementation.

Following the HLD, we move into the LLD phase, which focuses on the specific details of each module within the system. This phase includes defining the logic algorithms, data structures and interfaces required for implementation, serving as a blueprint for developers. At the same time, Yi Ya creates prototypes to enhance customer interactions by Figma, while Yoke Yu uses MySQL to develop a schema that effectively manages data related to users, applications and financial reports. By the end of this, we aim to deliver a set of documentation, including system architecture diagram and UI prototypes as a guideline to proceed into implementation.

Phase 3: Implementation (Duration: 5 weeks)

This phase begins by setting up the development environment and establishing version control to manage code changes and collaboration effectively. Jie Ying focuses on building the server-side logic for key features such as member registration, loan management, salary deduction and financial reporting. Simultaneously, Ru Qian is responsible for designing and coding the user interface using technologies like HTML, CSS, JavaScript and PHP to ensure a seamless user experience.

Once both sides are developed, our teams work together to integrate the front-end with back-end APIs, ensuring that the system's front-end can communicate with the back-end services for functions like displaying member profile management, loan repayment tracking and financial reporting. At the same time, Yoke Yu connects the application to the database to ensure data persistence and managing the storage.

By the end of this phase, the key deliverables include working modules for member registration, loan management, a fully functioning database and successful integration of front-end and back-end systems.

Phase 4: Testing (Duration: 2 weeks)

The purpose of this phase is to ensure the system's accuracy, functioning and general user satisfaction. In order to ensure that each module functions effectively, this step starts with unit testing. For instance, in the Member Registration Module, checks will be implemented to prevent duplicate accounts, validate input forms and enforce mandatory data. The Log In Module will be evaluated for proper login behavior, including scenarios for password resets, session management, and the proper display of invalid credentials. Tests will be conducted on the Loan Application Module to ensure that valid applications can be submitted, statuses are shown and duplicate or incomplete applications are prohibited. In the Financial Report Module, tests will verify that reports are generated correctly for valid date ranges and that invalid date entries result in error messages. Tests in the User Profile Management Module will ensure that users may update profile details, upload images and change passwords while obeying input validity and format requirements.

Following unit testing, Integration Testing will check that individual modules such as member registration, loan management, and financial reporting work together seamlessly, resulting in a consistent user experience across all features. After integration is complete, System Testing will evaluate the system's functioning in real-world scenarios based on end-to-end user interactions under different scenarios to ensure that the system functions as planned. This phase's deliverables will include complete test results, bug reports and user feedback documents that will help refine the system before final deployment.

Project Schedule

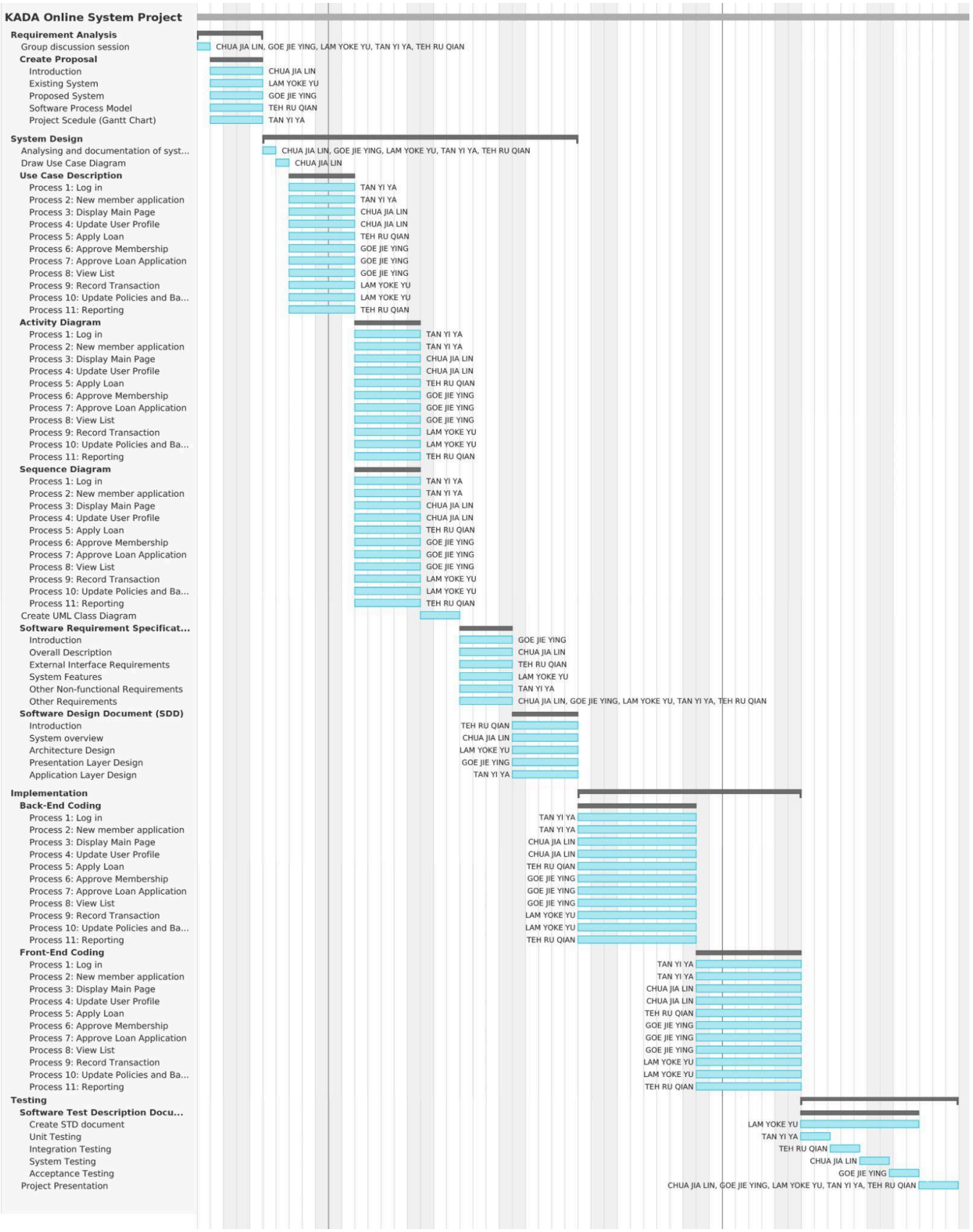



Diagram 2.2 Project Schedule